BT102(Microbiology)-MID TERM FILE -2022

ADMIRE GROUP OF ZOOLOGISTS



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SHORT QUESTIONS

1. Function of cell membrane factors of bacterial cell death? 5

cell membrane is composed of a phospholipid. molecules arranged tail to tail with heads facing away from each other. This makes lipid molecules to appear as a bilayer. So all membranes enclosing the organelle in the cell or the cell itself look as bilayer structures. This is the basic design of the plasma membrane. Protein and carbohydrates are also inserted into the plasma membrane.

There are some basic functions of cell membrane in bacteria.

1) Selective permeable barrier
2) Passive and Active Transport
3) Respiration in microbes (bacteria)
4) Photosynthesis in microbes
5) Lipid synthesis
6) regulates the transport of substances in and out of the cell.
7) receives chemical messengers from other cell.
8) cell mobility, secretions, and absorptions of substances

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• Cell wall parts are transported by a molecule called bactoprenol which is present in the plasmamembrane.

• PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc.

2. Oxidation phosphorylation and mechanism of phosphorylation ? 3

Oxidative Phosphorylation

a. Electrons captured from foods are transferred to co-enzymes such as NAD+ or FAD etc.

Then these electrons pass through a series of electron carriers and then ultimately to the last electron acceptor molecules such as O2 or other inorganic compounds (nitrate, sulphate, carbonateetc) through a series of these electron carriers in system called electron transport chains.

During these events, ATP is generated by chemiosmosis.

B. PHOTOPHOSPHORYLATION

This occurs in plants and chlorophyll containing microbes or photosynthetic cells. In these cells, lightenergy is finally converted into ATP. An electron transport chain is also involved here. This is summarized in the accompanying diagram above.

3. Dry heat method sterilization? 3

a) Form of dry heat include:
b) Dry heat (hot-air oven),
c) Flaming (Platinum loop sterilization in the flame of Bunsen burner),
d) Incineration (burning to ashes).
e) Similarly, moist heat can also be grouped into autoclaving (15psi), boiling at 100oC, and Tyndallization(100oC for 3 consecutive days to sterilize sugar solution that can be degraded by autoclaving) and pasteurization (usually at low temperature than 100oC).

4. F PLUS cell ?? 2

In E. coli, fertility factor (F factor) plasmid was the first plasmid observed to be transferred from one organism to the other, hence those bacteria that have this plasmid are called F+ cells to differentiate fromthose that do not have one (F-). However, once F- cells acquire F+ plasmid, they also become F+.

5. Western blotting explain? Most important 5 marks

Western Blotting: We can use this technique for the detection of antigen in the serum. Microbial proteins can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) is present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies as seen in the accompanying figure.

6. Function of cell wall of bacteria? 5

a) Cell wall is the most important layer in bacteria. It is rigid layer just outside the plasma membrane.Most important structure in prokaryotes
b) It provides shape to the organism.
c) It provides protection from osmotic lysis.
d) It is involved in pathogenicity (ability of the organism to cause the disease).Several antibiotics act on it.
e) Each bacterium is enclosed by a rigid cell wall composed of peptidoglycan, a protein-sugar (polysaccharide) molecule. The wall gives the cell its shape and surrounds the cytoplasmic membrane,protecting it from the environment. It also helps to anchor appendages like the pili and flagella, which originate in the cytoplasm membrane and protrude through the wall to the outside. The strength of the wall is responsible for keeping the cell from bursting when there are large differences in osmotic pressure between the cytoplasm and the environment.

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7. What is flagellum? And gives its function.

Long filamentous structure that propels bacteria is called flagellum. It helps bacteria to move.

8. Name two basic dyes?

In basic dye the color is in the positive ion. e.g

I. 1-Crystal violet
II. 2-Methylene
III. 3-blue Malachite green
IV. 4-Safranin

9. What is vertical horizontal gene transfer and write type of lateral gene transfer ? 3

Vertical Gene Transfer

This is a normal way of transferring genes from parents to offspring. This happens when a celldivides. Each daughter cell receives exactly what its parent cell has.

HORIZONTAL GENE TRANSFER

When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:

1. Transformation
2. Conjugation
3. Transduction

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10. Define osmotic pressure, osmotic pressure and plasmolysis?

Osmotic pressure is the force caused by a solution passing through a semi permeable surfaceby **osmosis**, whichis equal to the force required to resist the solution from passing back through the surface. An example of **osmoticpressure** is the process to filter water.

plasmolysis

If cells are placed in hypertonic solutions, water leaves the cells shrinking the cells and damaging them. This process is called plasmolysis.
Plasmolysis is the process in which cells lose water in a hypertonic solution

11. What is sporogenesis?

Formation of spores takes place within a vegetative cell and the process is called sporulation or sporogenesis. It is initiated when nutrients become unavailable.

12. Why 70% alcohol is more effective than 100% alcohol?

Ethanol, isopropanol are examples. Alcohols denature proteins, and dissolve lipids. Alcohols require water for being more effective. This is the reason that seventy percent alcohol is more effective than 100%. Please note that alcohols can effectively kill vegetative form of bacteria, and fungi but not spores. Alcohols are not very effective on wounds. Commonly used in hand sanitizers.

13. L form bacteria ??

Some cells, called **L-form** organisms, are naturally found without the cell wall. In other words, normally, they have cell wall but under certain conditions, they may lack a cell wall. These cells can be created when grown in the presence of penicillin which inhibits cell wall synthesis.

14. What is HRF and its functions??

If F factor gets integrated into the chromosome of the recipient cells, the bacterium becomes high frequencyof recombination cell (Hfr).

a. Conjugation can be used for gene mapping on the chromosomes.
b. conjugation_mapping.html
c. High frequency of recombination is also a kind of conjugation.

Can be used for mapping genes.

high frequency of recombination

In some cells that carry F+, F+ (Fertility factor) gets incorporated in the chromosomal DNA which converts F+ cells to high frequency of recombination cells (Hfr cells). When conjugation occurs between an Hfr cell and an F- cell, the Hfr cell's chromosome (with its integrated F factor) replicates,and a parental strand of the chromosome is transferred to the recipient cell.

15. Difference between viable and direct count ??

Viable count:

Viable count is a method used in cell culture to determine the number of living cells in a culture. This is different from other cell counting techniques because it makes a distinction between live and dead cells

Direct count:

Direct counting methods include microscopic counts using a hemocytometer or a counting chamber.The hemocytometer works by creating a volumetric grid divided into differently sized cubes for accurately counting the number of particles in a cube and calculating the concentration of the entire sample.

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16. Pilli and its functions??

These are hair-like structure composed of pilin, usually one to ten in number.

a. Longer than fimbriae
b. Used for attachment to:
   - Host cells
   - Bacteria
c. Used for DNA transfer from on bacterium to another:Conjugation (Sex pili)
d. Also function in twitching Motility

Gliding Motility is also the function of the pili.

17. Fermentation of fermentation??

Fermentation: Initial steps for glucose oxidation by fermentation are the same as they occur in glycolysisfor respiration. However, when pyruvate is generated through glycolysis, electrons are

also captured by NADH and this NADH needs to be regenerated into NAD+ for recycling. If the final electron acceptor is one of the end products such as ethanol, lactic acid, acetic acid etc, the process is called fermentation.

Both anaerobic respiration and fermentation do not use oxygen during these processes, and during fermentation, there are no electron transport chain involved as you can see in the accompanying diagram.

18. Process of germination of spore??

Spore germinates when it finds a conducive environment for its growth. Germination has 3 phases or stages:

Activation: It prepares the spore for germination. Heat can activate the spore when appropriate moisture andnutrients are present in the environment.

Germination: Spore starts swelling and loosing its coats etc. It becomes metabolically active.

Outgrowth: New components are made

19. What is magnestome ??

Inclusions of iron oxide

Surrounded by invaginations of plasma

membranePresent in G negative bacteria

Act like a magnet. Bacteria can stick to iron containing rocks for nutrition.

Decompose H2O2 which is toxic for cells.

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20. Bacterial growth curves phase ? most imp

There are four distinct phases of this curve.

1. **The Lag Phase:** cells prepares for growth in this phase. No growth is observed during this period or phase,however. Cells are metabolically very active during this phase.
2. **The Log Phase:** During this phase, organisms multiply exponentially or logarithmically. Generation time becomes constant during this phase and that is the reason, the log graph will show a straight line. Cells are in the most active stage during this phase of growth curve. For commercial applications such as vaccine production, cells have to remain in this phase in order to reproduce most efficiently resulting in increased cellmass or number. Another application of this phase of growth is to determine the generation time.
3. **The Stationary Phase:** This is also called a period of equilibrium as microbial deaths equal production ofnew cells. In other words, organisms start dying during this phase, however, the number of dead organisms isreplaced by new organisms because there is still replication of cells going on. So, overall number of organisms does not change. This is the reason, it is called a stationary phase.
4. **The Death Phase:** The number of deaths exceeds the number of new cells formed during this phase. In other words, overall number decreases. It is also called logarithmic decline phase. Why there is a decline phase or deathphase. The reason is simple: Nutrients are depleted and waste products which are toxic to the cells accumulate suppressing the growth and killing the cells.

21. Name five kingdom classification.5 marks.

22. Taxonomic hierarchy?

Microbes are placed in groups based on similarities that they share with each other. All organisms can be grouped into a series of subdivisions that make up the taxonomic hierarchy. A bacterial species represents —a monophyletic and gnomically coherent cluster of individual organisms that show a high degree of overall similarity with respect to many independent characteristics, and is diagnosable by a discriminative phenotypic property‖ (definition taken from the internet).

A genus consists of various specie; however, these species differ from each other in certain ways, although these are related by descent with each other. Related genera make up a family. A group of similar families constitutes an order, and a group of similar orders makes up a class. Related classes, in turn,make up a phylum. All phyla that are related to each other make up a kingdom, and related kingdoms are grouped into a domain. (see handouts)

23. Inclusion body define? 3

Inclusions are reserved deposits of nutrients. Bacteria accumulate nutrients when nutrients are plentiful. Concentrating nutrients in the form of inclusions avoids the increase in osmotic pressure due to accumulation of nutrients in one place. Not all organisms accumulate them. So, inclusions vary from bacteriato bacteria. They can also serve as markers for bacterial identification as some are limited to specific organisms. The followings are some of the important inclusions:

a) Metachromatic granules
b) Polysaccharide granules
c) Lipid inclusions
d) Sulfur granules etc.

24. Antisepsis, sepsis and sterilization ?

Antisepsis: Removing pathogens from living tissue

Microbial contamination A toxic inflammatory condition arising from the spread of an organism from a focus of infection.

Sanitization: Lowering microbial counts on eating utensils.

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25. Define oxidation reduction reaction?

Oxidation = loss of electrons, or gain of oxygen, or loss of hydrogen .

Reduction = gain of electrons, or loss of oxygen, or gain of hydrogen

26. Three characters of algae?

a. These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lacktissues such as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth. Microscopic exam is needed to identify unicellular and filamentous algae. However, multicellular algae that are commonly known as seaweeds are macroscopic in nature and can be identified morphologically without the help of a microscope.
b. Four groups of such algae include blue-green algae, green algae, brown algae and red algae. These algae are located in the sea at various locations and absorb light of various wavelengths, hence red algae are located far from the surface and can use blue light from the sun as blue light is of shorter wavelength and can penetrate deep in the sea. Please see the accompanying figure (above) for a better understanding of how wavelength relates with various algae.
c. Also remember, blue-green algae need magnification in order to be correctly identified, although they are not microscopic. o Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stemlike hollow stipes

and leaflike blades. There is no vascular tissue in these algae. Also, the stipe is not lignified or woody, so it does not provide support to the weed. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside themwhich keeps them floating in the water. This gas filled structure is called pneumatocyst or float.

d. These are eukaryotes. Their cell wall consists of cellulose. These are photosynthetic and produce oxygen. They are usually unicellular, but multicellular algae are also common. Seaweeds and pond scum are some ofthe examples.

27. Thermal death rate and thermal death point ??

Thermal Death Point: It is the lowest temperature at which all cells in a culture are killed in 10 min. Itwill be a specific temperature for a specific species of organism.

Thermal Death Time: Time (minimum) during which all cells in a culture are killed at a given temperature.This will vary from temperature to temperature for the same organism.
Obviously, higher temperatures will take less time to kill the organisms than low temperatures.

28. How bacteria can be identified? most imp

Microbes especially bacteria (disease causing) can be identified by three methods:

A. Classical or Conventional Method
B. Serological Methods
C. Nucleic Acid based Methods

Classical methods of microbial identification involve differential staining of the sample before it its culturing, culturing the sample onto nutrient agar, blood agar and MacConkey's agar, purifying the culture(colonies expected to be involved in the disease) and detection of various enzymes that belong to various metabolic pathways. Classically, such methods used
to take a long time to perform (3 days at least); however, rapid identification methods have become availablenow which use preformed media that can be used for testing the presence of metabolic enzymes. Such a kit isshown in the figure above (previous page).
Serological methods involve antibodies and antigen interactions. Antibodies are produced by B lymphocytesagainst any foreign antigens such as bacteria and their toxins that enter the body of animals or humans.
Antibodies are very specific in their interaction. In other words, antibodies made against E. coli do not bind or interact with Staphylococci or vice versa. Although, there are many serological techniques that can be successfully used for identification of microbial infections, we will focus on only a few of them.
The main advantage of using serological method is the speed and economy. In other words, serological methods are quick to do (take only about an hour) and can work directly on the sample (sample does not have to be cultured and purified as is needed in conventional methods of identification). Serological methods can be performed on cultured microbes as well which again speed up the diagnosis. Also remember that in all theseserological tests, either the serum or the antigen should be known to us.

29. Catalase and its protocol??

When H_2O_2 is produced, it gives rise to peroxide anions which are toxic to the cells. Cells have evolvedto deal with this toxic H_2O_2 by an enzyme called catalase as under:

Reduction = gain of electrons, or Loss of oxygen, or gain of hydrogen
Oxidation: Removal of electrons from an atom

• **Reduction**: Gain of electron • Oxidation and reduction reactions are coupled.

30. Five kingdom classification? 5 imp q

In 1969, five kingdom classification was proposed by Robert Whittaker as under:

1. Monera: Bacteria
2. Plantae: plants
3. Animalia: Animals
4. Fungi: Yeasts, molds and mushrooms
5. Protista: These are unicellular eukaryotes. Organisms that do not fit into any other category are placed inProtista. They are larger than prokaryotes. They include algae, protozoa, slime molds and water molds.

31. Using ionization radiations?

include gamma rays, X rays, or high-energy electron beams. The ionizing radiation possesses a wavelength
shorter than that of nonionizing radiation, less than about 1 nm. All these short wavelengths cause ionizationof water generating high reactive hydroxyl radicals which are damaging to the cells as they bind to DNA andresults in mutations. Medical plastic supplies, medicines and meat products can be sterilized by radiation.

32. Three characteristics of integral protein?

integral proteins are not static in position in the membrane. They can diffuse laterally and change positions in the membrane from time to time. So, plasma membrane is like water pond and integral proteins are like plastic bags people throw in the pond. These plastic bags swim and move by air currents. Membranes are much like that. Integral proteins act as channels or carriers in facilitated diffusion. Integral proteins are called transporters or permeases.

33. Definition of gene, chromosome genome, phenotype 5 marks

34. Identification methods for microbial growth 5 marks

35. Pili and its types5 marks. See handouts

36. Differ between oxidation and reduction. See google

37. Buffer explain see handouts

38. Name of bactaria which can grow on optimal temperature ?

Organisms are basically classified into three groups based on the temp requirements:

Psychrophiles: These are further divided into strict psychrophiles and psychrotrophs:o **Psychrotrophs:** Cold loving: 15 0C

a. **Psychrotrophs:** Optimum temp is 20-30 0C. Food spoilage bacteria that can spoil food duringrefrigeration.

Mesophiles: 25 – 40 0C: These are the ones that cause diseases in animals and humans.

b. Moderate temp loving organisms

c. Optimum: 37 0C as this is the body temperature of humans and animals.

BrThermophiles: 50 – 60 0C

d. Heat loving

e. Important in organic compost piles

f. These are further divided into hyperthermophiles that grow optimally at 80 0C. They live in hot springs.

39. Branches of mycology?

Mycology is the branch of biology concerned with the study of fungi, including their genetic and biochemical

properties, their taxonomy and their use to humans as a source for tinder, traditional medicine, food, andentheogens, as well as their dangers, such as toxicity or infection.

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- Medical mycology
- Agricultural mycology

Ecological mycology – Decomposers of woods, leaves, feces and other organic matters.

40. Difference between precipitation and **agglutination**?

a) Agglutination Test: Let's first talk about a few definitions that will make our life easier. There are two
b) words we must know about them: They are agglutination and precipitation. Agglutination is a process inwhich soluble antibodies interact with a particulate antigen. Precipitation is a process in which soluble antibodies interact with a soluble antigen. Now let's talk about a few examples of agglutination tests:
    a. Blood Grouping: This is a typical agglutination test which involves antibodies (that are always solublein nature) and RBCs (that are particulate in nature).
    b. Rose Bengal Plate Agglutination Test: This test is used to diagnose Brucellosis in animals in which serum of animals suspected for Brucellosis is mixed with Brucella antigen (bacterim). The resulting positive interaction is seen as flocculation as are shown in the accompanying figure:
    c. Widal Test: This test involves Salmonella cells (Antigen) and Salmonella antibodies.
    d. Coombs Test: This involves antibodies coated RBCs (antigen) and antiantibodies.
c) agglutination tests are qualitative tests used to detect the presence of antibodies in serology laboratories andblood banks. Treated red blood cells or colored latex beads, coated with antigen, clump in the presence of antibody to the antigen.
d) Precipitation Tests: These involve soluble antigens and soluble antibodies. Examples include:
e) Agar Gel Diffusion Test: This test is performed in agarose gel which is poured in a petri dish and wellsare created which are used to place antigens and antibodies for interactions with each other. Antigens and antibodies diffuse from their respective wells and, where these molecules meet each other, a precipitation line is formed as shown in the accompanying figure.
f) Precipitation of proteins: This method can also be used to isolate a specific
g) protein from cells by using protein specific antibodies which precipitate out that protein from a mixture ofcellular proteins.
h) A precipitation reaction is a test which is involved in the serology, for the detection of antigens and antibodies.
i) Precipitation test is used to quantify both antigen and antibody, and therefore its specificity also depends uponthe concentration of both the reactants

41. Conjugation, transformation, transduction?

Conjugation is the process by which one **bacterium** transfers genetic material to another through direct contact. During **conjugation**, one **bacterium** serves as the donor of the genetic material, and the otherserves as the recipient

Genes transfer process can be mediated by plasmids. Plasmids are extra chromosomal circular DNA fragments that replicate independently of bacterial chromosome. Conjugation requires direct cell to cellcontact and cells have to be opposite mating types.

Transduction is the process by which foreign DNA is introduced into a bacterial cell by a virus or viralvector. An example is the viral transfer of DNA from one bacterium to another and hence an example ofhorizontal gene transfer

Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When abacteriophage infects a bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is called a capsid) in which viral DNA is packed before the virus is released.

Bacterial transformation is a process of horizontal gene transfer by which some **bacteria** take upforeign genetic material (naked DNA) from the environment ..... The prerequisite for **bacteria** to

undergo **transformation** is its ability to take up free, extracellular genetic material. Such **bacteria** aretermed as competent cells.

Naked DNA in solution is transferred from one bacterium to another during this process.

42. Antibacterial chemical?

Although antibiotics are used to kill organisms in disease states, some antibiotics are not very effective for this purpose; however, these antibiotics could be used in food to prevent food spoilage. Nisin and natamycin prevent spoilage of cheese.

43. Define culture, medium ,inoculum?

a) This term is used to define growth of microbes. In other words, microbial growth in the lab is called aculture.

b) A nutrient material that supports the growth of microbes in the lab is called a culture medium.A culture media is a special medium used in microbiological laboratories to grow different kinds of microorganisms. A growth or a culture medium is composed of different nutrients that are essential for microbial growth.

c) **Inoculum** Microbes introduced into a culture medium that initiate growth of organisms.
inoculum A small amount of material containing **bacteria**, viruses, or other microorganisms that is used to start aculture.

44. Advantages of staining?

a) Most organisms appear colorless when seen under a microscope.
b) Staining emphasizes certain structures of the organisms.
c) Staining is just coloring with a dye.
d) Staining increases visibility of microbes because staining increases contrast.
e) Shape, size and arrangements of the organisms can be readily seen.
f) Purity or contamination of a culture could be determined.
g) Differentiation and classification of microbes is possible. For example, microbes can be categorizedinto Gram positive or Gram negative groups.
h) Structures such as flagella, capsule and spores etc. of bacteria can be detected with staining.

a. It gives quick results when examining infections.
b. It is simple and cost-effective.
c. It helps with determining appropriate treatments for infection.
d. It allows for various methods of testing.
e. It is basically a key procedure in identifying bacteria.

45. Five layers of endospore?

a. **Exosporium:** A thin delicate outermost covering of the spore
b. **Spore coat:** 2nd layer underneath the exosporium. It is thick and composed of several protein layers.Resistant to chemicals .It contains enzymes for germination. Germination of spores into vegetative form occurs when environment becomes favorable for their growth.
c. **Cortex:** It is the 3rd layer from outside in. It has peptidoglycan in it.

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d. **Spore cell wall or core wall:** Surrounds the protoplast or spore core
e. **Spore core:** Contains nucleoid and ribosomes

46. Name three buffer for controlling ph in culture medium?

a) Peptones
b) amino acids
c) PBS

47. Three shapes of bacteria?

Coccus These are circular or spherical in shape.

a. Bacillus These rod shaped bacteria.
b. Spiral These are curved shaped bacteria. They are further divided into 3 moresubgroups.

Vibrio: curved rods

Spirillum: Helical but
rigid Spirochete: Helical
but **flexible**

c. Spirochetes move by axial filaments which are enclosed by the outer membrane.

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MCQ BT-102 – MICROBIOLOGY - 2022

1. Why do we Classify Organisms? (Easy Identification).
2. Five Kingdom Classification of Organisms (carl Linnaeus 1735).
3. Three Domain Classification given by (1977, carl woese)
4. Three domain classification include (Bacteria: True or eubacteria , Archea, Eucarya)
5. Bactaria are (Procaryotes, Usually single-celled organisms, Cell wall made up of, peptidoglycan).
6. Archea
7. Live in extreme conditions (Thermophiles o Halophiles).
8. ➢Procaryotes are (Cell wall has no peptidoglycan, but has special lipids, Vary in rRNA sequencing).
9. Live in extreme conditions (Thermophiles Halophiles).
10. Eukarya include (Protists, Fungi, Animals, plants).
11. Protists include (Algea, protozoa, slim moldes).
12. Fungi Unicellular form is called(yeast)
13. Multicellular is known as(mold or mushroom)
14. Viroids: composed only of circular ssRNA (potato spindle tuber viroid).
15. Virusoids: Contain circular ssRNA, need helper viruses for replication and encapsidation; also called (satellite viruses)
16. Escherichia coli honors the discoverer, Theodor Escherich describes the

bacterium‘s habitat (the large intestine, or colon).

17. Escherichia coli and Staphylococcus aureus are found in the human body (E. coli is found in the large intestine, and S. aureus is on skin).
18. First microscopic observation of microorganism (Lucretius, Fracastoro).
19. First published drawing of microbes in (1665, Robert Hooke).
20. First to publish extensive and accurate observations of,microbes. He called them —animalcules‖ (Antony van Leeuwenhoek).
21. Who Conclusion organic matter has vital force (John Needham ,1748).
22. Who applied phenol as disinfectants in treating surgical wounds and hand sanitizing(1860, Joseph Lister).
23. In 1865, Pasteur found that silkworm disease caused by a protozoan.
24. In the presence of O2, bacteria grow and change alcohol to acetic acid said by (pasture).
25. Germ Theory of Disease given by (pasture).
26. Who applied phenol as disinfectants in treating surgical wounds and hand sanitizing (1860, Joseph Lister).
27. Pasteur found that silkworm disease caused by a protozoan in (1865).
28. In 1876 Bacillus anthracis established as a cause of anthrax first proof given by (Robert Koch).
29. Vaccines discovered by Edward Jenner in 1796.

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30. Pasteur in 1888, discovered why vaccination works Avirulent microbes as vaccine.
31. Golden age of microbiology include (1857 to 1911).
32. Medical Microbiology Deals with diseases of humans and animals.
33. Epidemiology Monitoring diseases in a population.
34. Industrial Microbiology Use of microbes or their enzymes for large scale production of biomolecules.
35. Illuminator used as a light source.
36. Condenser used to Directs the light through the specimen.
37. Objective Lenses used to Close to the specimen.
38. Ocular Lens (Eyepiece) which is Close to the eye.
39. Objective lenses: Low: 10 , High: 40 X, Oil immersion: 100 X.
40. What is resolution? Ability of the lenses to distinguish between two closely lying objects as separate
41. Light microscope resolving power is 0.2 µm.
42. Darkfield Microscopy Uses a darkfield condenser having an opaque disk.
43. Fluorescent Microscopy Uses fluorochoromes and UV light as illumination source.
44. Fluorochromes: substances that absorb short wavelength of light and emit
45. longer wavelenght (visible).
46. Advantages of Staining, Increases visibility of microbes, Shape, size and arrangements, Purity or contamination of a culture, Differentiation and classification of microbes, Various parts of bacteria can be detected with staining.
47. The specimen is spread into a thin film (smear).
48. Bacteria are negatively charged at pH 7.
49. Basic dye: The color is in the positive ion.
50. Acidic dye: The color is in the negative ion.
51. Basic Dyes: Positive Staining Crystal violet, Methylene blue, Malachite gree, Safranin.
52. Acidic Dyes: Negative Staining ,Eosin Acid, fuchsin.
53. Most common is Gram staining 1884, by Hans Christian Gram.
54. Which of the following contains mycolic acid? Mycobacterium and Nocordi
55. After gram staining, gram positive bacteria appear.purple
56. Rod shaped bacteria are called. bacilli
57. Energy is required in. activemovement
58. Which of the following structure is involved in the locomotion of bacteria? flagella
59. Gram positive bacteria produced exotoxins.
60. The basal body of flagellum of gram positive bacteria consist of rings.
61. 70s ribosome consist of two subunits. 50S and 30s
62. 9. Are used for twitching motility. pili
63. Integral protein are imbedded in the cell membrane and are insoluble in water.

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64. Mostly formed by Gram positive bacteria Clostridium, Bacillus
65. One gram negative: Coxiella burnetii also have it.
66. Formation of spores is called sporulation or sporogenesis.
67. Spore core: Contains nucleoid and ribosomes.
68. Which of the following is the cellular energy of most of the microbes , All of these.
69. Which of the following is used for catalase test hydrogen peroxide.
70. In anaerobic respiration, final electron acceptor is in organic substance other than O2 is all of the given.
71. Microorganisms that have high nutritional requirments are known as fastidious.
72. Which of the following is acidophile both molds and yeast.
73. Bactaria gets sulfer from all of the given.
74. Which of the following staining technique is used for mayobaterium species none of the given species.
75. Which of the following is the moist heat sterilization all of the given species.
76. Aerobes that require o2 in low amounts is called microaerophiles.
77. The cell wall of archea composed of speciallipids.
78. Which is the correct order of acid fast staining carbol fuschin>crystal violet >decolizer >methylene blue.
79. Photoautotrophs use as a source of energy sunlight
80. Lipid bilayer is present in cell membrane.
81. A nutrient material for growth is known as culture medium.
82. Prokaryotes have no cellmembrane.
83. Which of the following is used for catalase tets hydrogen peroxide.
84. Bacteria need for synthesis of nuceic acid sulpher and phosphores.
85. Chemoautotrophs use carbonfrom Co2
86. Cell wall of algea made up of cellulose .
87. Which of the following is physical method of microbial control all of the given.
88. The alcohol ethanol, isopropanol are used for the denaturation of protein.
89. Energy is required in activemovement.
90. Boiled extracts of hay or meat can give rise to microorganism this statement proposed by johnneedham.
91. When to round shape bactaria join the its known as diplococci.
92. Sepsis is states as microbial contamination.
93. Microbes are placed in monera Protista and fungi.
94. Which of the following staining is done for acid fast bactaria ziehl nelson staining.
95. Which of the following is the function of cell wall all of the given.
96. Anthrax is caused by bacillus anthracis.
97. Which of the following is required for treatment of disease and cancer antibodies.
98. Dye of negative staining eosin.
99. All organic compound contains carbons.

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100. Which of the following staining techniques used for mayobactarium species none of the given option.
101. Kingdom monera both include archea and bactaria.
102. Soluble antigen binds with antibodies in known as precipitation test.
103. Which of the following gas is used for microbial removal Ethylene oxide.
104. Which of the following the bacterial gene transfer method transfection.
105. Which of the following compound are needed for protein synthesis nitrogen and sulpher.
106. The oil immersion lens of microscope is of 100X.
107. Extrachrmosomal DNA is known as plasmid.
108. There are 5 kingdomin Eukarya.
109. Cold loving bactaria is known as phycophiles.
110. Function of cell wall all of given.
111. Organisms are classified into 5 kingdom in latest classification.
112. Hair like small appandages on gram negative cells are known as Fimbrae.
113. Salts composed of positive and negative ions known as stains.
114. Prokaryotes do not have distinct nuclear membrane.
115. Phychrophiles are extremophile oraganisms that are capable of growth and reproduction in temperature around 15 centograde.
116. Branch of scienece deals with the bactaria are called bacteriology.
117. Are mostly in extreme condition archea.
118. Which of the staining technique is used for myobactarium species none of the given species.
119. Prokaryotes have no all of the given option.
120. Which of the stain used for simple stain methylene blue.
121. The habitat of E. coli is larg intestine.
122. Which of the stain co tains myolic acid myobactarium and nocordia.
123. Which of the following is the dye for positive staining all of the given.
124. Chemoautotrophs use carbon from carbohydrates
125. Which rate is constant for microbes growth rate.
126. Which of the following used for antimicrobial agent.
127. Naming of culture introduce by Linnaeus.
128. After gram staining gram staining bacteria appear purple.
129. Population of one organisms derived from the other is called clone.
130. The basal body of flagellum of gram positive bactaria consist of 2 rings.
131. Which of the following used for gene mapping conjugation.
132. Monitoring of disease in population is known as epidemiology.
133. Oxygen is toxic for Microaerphile.
134. Which is the correct answer about sterilization removal of microbes and their spores.

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135. Which of the following is correct about myobactarium tuberculosis obligate aerobes.
136. Which of the following used for disease and cancer antibiotics .
137. Which of the following has high susceptibility to to pencilin gram positive bactaria.
138. Which of the following used for attachment both fimbrae and pili.
139. Used as antimicrobial agent all of the give .
140. Which of the following need for ATP synthesis nitrogen and phosphores.
141. Which of the reason for changing in classification all of these.
142. Bactaria obtain nitrogen from atmosphere by nitrogen fixation.
143. The man behind cell theory is Robert hook.
144. Which of the following is not a bacterial gene transfer methid transfection.
145. Spirochete bactaria move by axial filaments.
146. The primary stain of gram methods crystal violet.
147. Which of the following used for attachment both fimrae and pili.
148. Which of the following required to see microorganisms all of these.
149. Microbes first drawing published in1665.
150. Removing of microbes from a limited area is known as degeming.
151. Flagella is responsible for locomotion
152. Which of the following enzyme catalyze the conversion of hydrogen peroxide into water and oxygen catalase.
153. Oxygen is toxic for microaerophiles.
154. The study of immune system is called immunology.
155. Cytoplasm contains 80% of water.
156. Which of the following used for microbial removal ethylene oxide.
157. Which stain used for acid fast bacteria ziel nelson staining.
158. Which of the following is the cellula energy for most of the microbes allof these.
159. Which of the following is used for catalase tets hydrogen peroxide.
160. The domain classification purposed by carl woese.
161. In electron respiration, finals electron acceptor is electronic substance other than O2 is all of the given
162. Bacteria get nitrogen from all of the given.
163. NADH and FADH2 are oxidizing during electron transport chain.
164. Gram staining fall into which of the categories differential staining.
165. Which of the following formed inside the bacterium endospore.
166. Antharx is caused by bacillus anthracis.
167. Who discovered the first vaccine Edward jenner.
168. Which of the follwoeing relevant to respiration of glucose all of the given option.

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169. Which of the following is not present in bactaria both histones and nuclear membrane
170. we can we fix the smear both physically and chemically.
171. What are the physical requirement of bactaria all of the given option .
172. Lipid bilayer is present in cell membrane.
173. Faceplate diffusion is donewith the help of intergral protein.
174. Which of the is responsible for destruction of cell membrane all of the given option.
175. Which of the following bactaria has waxy materials in the inner cell wall myobactarium tuberculosis.
176. Which of the following is the illumation is the source for flourecent microscope both flourochomes and vu lights .
177. Half the dry weight of the microbes is carbons.
178. Motile and unicellular organisms these characteristics related to which of the following protozoa.
179. Bacteria utilize oxygen for aerobic respiration
180. Facilitated diffusion is one done with the help of Integral proteins.
181. Which of the following used for cell membrane destruction all of the given.
182. Which bactaria have waxy material in their cell wall mycobacterium tuberclosis .
183. Which of the following is the illumination source for flourecent light both flurochrome and UV light.
184. Half the dry weight of microbes is carbons.
185. Motile and unicellular organisms these characteristics related to which of the following protozoa.
186. Bactaria utilize oxygen for aerobic respiration.
187. Bactaria need nitrogen for the synthesis of protein and nucleic acid.
188. Which of the following is not present in bacteria both histones and nuclear membrane.
189. All organic compounds contains carbons.
190. Which of the following used as anti microbial agent all of the given.
191. Microoragisms belonging to the Aerotolerent aerobes.
192. Bactaria gets nitrogen from all of the given.
193. Pili are used for twitchingmovement.
194. Which of the following used as myolic acid myobactarium and nocordia.
195. NADH and FADH2 are oxidizing during electron transport chain.
196. Growth of a medium in a liquid media with closed vessel is known as batch culture .
197. Removal of electron from an atom is oxidation.
198. Too small to be seen with unaided eye both germs and microorganisms .

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199. Spirochete bacatria move by axial filament
200. Falagella is responsible for locomotion.
201. Sepsis is stated as microbial contamination.
202. Which of the following used for DNA transfer pili.
203. The gram stain of grams methods is crystal violet.
204. Which of the following is the smallest bacteria mycoplasma.
205. Transduction is the transfer of the DNA by a virus called bacteriophage.
206. Buffer are used to maintain the ph of a medium.
207. Microbes are placed in the monera Protista and fungi kingdom.
208. Monitoring of a disease in a population is called Epidemiology.
209. Pasting of a sample on a glass slide in known as smera formation.
210. Microscopy that Is required for unstained specimen is known as Dark field microscopy .
211. Prokaryotes have no all of the given
212. O2 has no effect on aerotolent aerobes.
213. Which of the following is the enumeration of live bataria cell all of the given option.
214. The habitat of e coli is large intestine.
215. Which is the physical source of growth all of the given
216. Antigen antibody reaction is known as Agglutination.
217. Which of the following ph is acidic all of the given
218. Which of the following is the dye of negative staining Eosin.
219. Pili is composed of pilin.
220. Which of the following is the first living organisms of our planet bacteria
221. Psychrophiles are extremophilic organisms that are capable of growth and reproductionin temratute around 15centgrade.
222. For smear prepration we use platinum loop.
223. Rigid membrane outside the plasma membrane is cell wall.
224. The cell wall of bacteria is motely made up of peptidoglycan.
225. Which of the following structure is used for locomotion in the bactaria flagella.
226. Which of the following is used for inhibition of protein synthsis both iodine and chlorine .
227. Photoautotrophs used carbon from CO2.
228. Phenol and phenolics compound first widely used antiseptics for disruption plasma membrane.
229. Which of the following is relevant to respiration of glucose all of the given option.
230. 70S ribosome consist of two subunits 50s and 30S.
231. The halogens iodine and chlorine inhibit protein function.

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232. Prokaryotes and eukaryotes are chemically similar .

233. Halicobactor pylori is aerotolerantaerobe.

234. Which of the following initiate the spore formation Heating.

235. Besides water ----------is the most important requirmen for microbial growth . carbon.

236. Removalandgainofelectronfromanatomisknownas Redoxreaction.

237. Bactraiaobtainnotrogebfromtheairbytheprocessof nitrogenfaxation..

238. Which of the following group of microbes strictly need oxygen for growth…… Obligates Aerobes

239. Which of the following is required to see microorganisms ..... All of the given options.

240. A DNA segment which encodes protein is known as .....Gene.

241. The alcohol ethanol isopropanol are used for the denaturation of.....Protein

242. After gram staining, Gram positive bacteria appear…… Purple.

243. Which of the following gas is used the microbial removal…..Ehylene oxide.

244. ……..work ended the spontaneous generation controversy.....Louis Pasteur.

245.8. Which of the following bacteria formed endospores….. All of the given option.

246. Salt composed of a positive and negative ions are known as…. Stains,

247. Motile and Unicellular organisms, these characteristics related to which of the following…… Protozoa.

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248. Oxygen is toxic for .....Obligate Anaerobes.

249. . Flagella move…….for forward thrust……Counter Clochwise.

250. Half the dry weight of microbes is ..... Carbon.

251. 70s ribosomes consists of two subunits..... 50s and 30s.

252. 15. There are…. Kingdom in domain Archea ..... 1.

253. 16. How we can fix the smear….. Both Physical and Chemical method.

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254.. The basal body of flagellum of gram positive bacteria consist of…..rings…. 2.

255. The theory of spontaneous generation was proposed by…. Aristotle.

256. Which of the following is correct about E. coli……Facultative Anaerobes.

257. Prokaryotes and Eukaryotes are ……similar….Chemically.

258. ……Are live mostly in extreme conditions…. Archea.

259. Substances that absorb short wavelength of light and emit longer wavelength are known as….. Fluorochromes.

260. Which of the following structure used for moving…. Flagella.

261. The oil immersion lense microscope is of……power… 100x.

262. First drawing of microbes was published by…….Robert Hooks.

263. Buffer are used to maintain…… in the medium…. PH.

264. Ability of the lens to distinguish between two closely lying objects as separate is known as……. none of the given option.

265. Photo autotrophs used…….. as a source of energy…… Light.

266. . Which of the following pH is acidic ……. all of the given option.

267. Rigid layer outside the plasma membrane is called……cell wall.

268. Bacteria utilise oxygen for…… decomposition

269. Phenol and phenolic compound first widely used antiseptics for disruption… plasma membrane.

270. Facilitated diffusion is done with the help of…… integral protein.

271. . Kingdom Monera includes both …… bacteria and Archea.

272. Photoautotrophs use carbon from… ……. CO_2.

273. Chemoheterotrophs use carbon from ……. carbohydrates

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274. Bacteria need ……. for the synthesis of protein and nucleic acids……Carbon.

275. Which of the following is used for DNA transfer…… Pili.

276. O2 has no effect on..... Aerotolerant Aerobes.

277. Removal of electrons from an atom is known as…… oxidation.

278. Which of the following is the physical method of microbial control…….
radiation

279. Which of the following gas is used for microbial removal… .....ethylene oxide

280. Naming of the organisms introduced by..... linnaeus.

281. ……. Are used for twitching motility…… pili.

282. Which of the following is the most important structure in prokaryotes…..
cell wall.

283. Rigid layer outside the plasma membrane is ..... cell wall.

284. Monitoring of diseases in population is known as…… Epidemiology.

285. Which of the following bacteria has waxy material in their cell wall…… Mycobacterium Tuberculosis.

286. Soluble antigen binds with antibody is is known as as…… Precipitation Test.

287. NADH and FADH2 during krebs cycle oxidized where electron transport chain in generate……. ATP molecule…..

288. Capsule is responsible for…..Virulence.

289. …. Is transfer of DNA by a virus called bacteriophages.....transduction.

290. Which of the following is used for plasma membrane is description…… Phenol.

291. Bacteria get nitrogen from… .....All of the given options.

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292. Which of the following staining is done for acid fast bacteria….. Ziehl Neelsonn Staining.

293. Which of the following is used for catalyst test…. Hydrogen peroxide.

294. Which of the following is the advantage of staining the microbes…… All of the given options.

295. All organic compounds contain…. CARBON.

296. . The multicellular form of Fungi is known as….. mold.

297. There are ..... kingdoms in Domain Eukarya…… 5.

298. Which of the following is not present in bacteria….. Both Histone and Nuclear membrane.

299. . Microbes are placed in ……kingdoms ..... Monera Protista and fungi.

300. Which of the following enzymes catalyse the conversion of hydrogen peroxide into water and oxygen… Catalase

301. The study of microbes is known as…… microbiology.

302. Psychrophiles are extremophilic organisms that are capable of growth and reproduction in temperature around ..... 25 to 40 centigrade.

303. Which of the following is hyperthermophile in nature..... Archae.

304. Which of the following is the cellular energy for most of the microbes All of the given options

305. Rigid layer outside the ..... plasma membrane is Cell wall

306. Capsule is responsible for ……..virulence

307. Which of the following is the function of cell wall?..... All of the given option

308. Lipid bilayer is present in

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____.....Cell membrane

309. Movement against the concentration gradient is known as .....Active transport

310. LPS stands for .....Lapopolysachrides 7

311. Cytoplasm contains wat .....80% percent

312. Which of the following has high susceptibility to penicillin?..... Gram Positive Bacteria

313. Which of the following contain mycolic acid?..... Mycobacterium & Nocardia

314. Which of the following is used for attachment?. .....Both fimbriae & pili

315. Which of the ..... following is the most important structure in prokaryotes?..... Cell Wall

316. Which of the following is the smallest bacterium?..... Mycoplasma

317. Flagella is responsible for Locomotion

318. Which of the following .... is used for DNA Transfer?. Pili

319. Flagella move_____for forward thrust..... Both counter clockwise& run/swim

_____are used for twitching motility... .....Pili

Which of the following is responsible for protein synthesis?. Ribosomes

320. Which of the fallowing is the type of membrane protein?..... Both peripheral &

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integral proteins

321. Which of the..... following bacteria has atypical cell wall?. Mycoplasma

322. After Gram staining, Gram positive bacteria appear..... Purple

323. Extrachromosomal DNA is known as. .....Plasmid

324. Gram Positive bacteria produce _____.....Exotoxins

325. Gram negative bacteria produce .....Exotoxins and Endotoxins

326. 70S ribosomes consist of two subunits,_____..... 50S, 30S

327. Rod shaped bacteria..... are called Bacilli

328. Which of the following . structure used for moving?. Flagella

329. Lipid A is responsible for .....All of the given option

330. Which of the following disease caused by spirochetes?. ..... All of the given option

331. _____ is composed of pilin ..... Both Pili & Fimbriae

332. . Which of the following is responsible for destruction of cell

333. membrane?.....All of the given options

334. . The basal body of flagellum of Gram-positive bacteria consist of _____ring.....2

335. . The basal body of flagellum of Gram-negative bacteria consist of _____ring... 4

336. Integral proteins are_____the cell membrane and are_____in

337. water.....

..... Imbedded in, insoluble

338. Hair like small appendages on G-neg. cells are known as Fimbriae

339. Which of the following is not present in bacteria?.... both histone & nuclear membrane

340. Facilitated diffusion is .....done with the help of Integral protein

341. Spirochetes bacteria .....move by Axial filaments

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342. Long filamentous structure that propels bacteria is known as Flagella

343. . Whn cocci join in chain like structure then the shape is known as .....Streptococci

344. . Which of the following structures is involved in the locomotion of bacteria Flagella

345. Which of the following condition in which water moves into the cell?. Hypotonic

346. Teichoic acid is made. up of All of the given option

347. Energy is required in…….. Active movement

348. Which of the following is used for mating?..... both Pili & Fimbriae

349. Which of the following is not part of the flagella?.....None of the given option

When two round shape bacteria join then it is known as ……. Diplococci

350. The oil immersion lens microscope is of _____ 351. Power..... X

352. Buffers are used to maintain _____In medium PH

353. First drawing of microbes was published by .....Robert Hooke

354. Photoautotrophs use energy..... Light as source of

355. Ability of the lenses to distinguish between two closely lying objects as separate is

356. known as ..... None of the given options

357. Which of following pH is acidic?..... All of the given options

358. Bacteria utilize oxygen for :..... Decomposition

359. Phenol and Phenolics compound First widely used antiseptics for 360. disruption..... .....Plasma membrane

361. Photoautotrophs use Carbon from CO2

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362. Kingdom Monera includes..... Both bacteria and Archae

363. . How we can fix the smear?..... Both physical and chemical method

364. . Bacteria need ____ acids for the synthesis of protein and nucleic Carbon

365. Chemoheterotrophs use.....Carbon from... Carbohydrates

366. Substances that absorb short wavelength of light and emit longer wavelength are

367. known as ..... Fluorochromes

368. O2 has no effect on Aerotolerant Aerobes

369. Salts composed of....a negative and positive ions are known as stains

370. Removal of electrons from.....an atom is known as Oxidation

371. Which of the..following is the physical method of microbial control?.....Radiation

372. Naming of the organisms .....introduced by Linnaeus

373. Which of the following gas is used for microbial removal?..... Ethylene oxide

374. "Motile and unicellular organisms", these characteristics related to which of the

375. following:..... .....Algae

376. . A DNA segment which encodes protein is known as ..... Gene

377. Oxygen is toxic forFacultative Anaerobes

378. _____Work ended the spontaneous generation controversy .....Louis Pasteur

379. . Which of the following group of microbes strictly need oxygen for 380. growth?..... Obligate Aerobes

381. There are _____ Kingdoms in domain Eucarya 4

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382. Half the dry weight of microbes is ____..... Carbon

383. The alcohols Ethanol, isopropanol are used for the denaturation of Protein

384. Which of the following is required to see microorganisms?..... All of the given options

385. Which of the following bacteria formed endospore?..... All of the given options

386. Monitoring of diseases in Populations bs Known as.....
Epidemiology

387. Which of the following bacteria has waxy material in their cell wall?..... Mycobacterium tuberculosis

388. Soluble antigen binds with antibodies is known as.....
All of the given options

389. NADH and FADH2 are oxidized during.....
Electron transport chain

390. _____ is transfer of DNA by a virus called bacteriophages ..... Transduction

391. Which of the following is used for plasma membrane disruption?..... Both phenol and bisphenol

392. Which of the following staining is done for Acid Fast Bacteria?....
Ziehl Neelson Staining

Which of the following is used for catalase test?.....
Hydrogen peroxide

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393. Which of the following is the advantage of staining the microbes?..... Increases visibility of microbes

394. All organic compounds contain _____.....Carbon

395. The unicellular form of fungi is known as .....Yeast

396. The multicellular form of fungi is known as .....Mold or Mushroom

397. In which of the following kingdom microbes are placed?..... All of the given options

398. Microbes are placed in _____ Kingdoms.....
Monera, Protista and fungi

399. Which of the following enzymes catalyze the conversion of atmospheric oxygen into hydrogen peroxide?. ..... Catalase

400. The study of microbes..... is known as Microbiology

401. Psychrophiles are extremophilic organisms that are capable of growth and

402. reproduction in temperatures around 1 centigrade

403. Which of the following is hyperthermophile in nature?. Archae

404. Which of the following is the cellular energy for most of the microbes?.....All of the given options

405. Which of the following is the enumeration of live bacteria cells?..... All of the given options

_____era is called golden age of microbiology

..... 1857 – 1914

406. Microorganisms belonging to the group, do not use O2 but can tolerate it fairly

407. well.....

..... Aerotolerant Aerobes

408. Mycology is the study of .....

None of the given options 100. The cell wall of Archea composed of special Lipids

409. Too small to be seen with the unaided eye..... Both germs & microorganisms

410. Bacteria get sulfur from .....

All of the given options

411. . Bacteria get nitrogen from.....

All of the given options

412. . Which of the following is required for aerobic respiration?.....

Hydrogen peroxide

413. Which rate is constant for microbes?..... None of the given options

414. The halogens lodine and chlorine Inhibit _____ function..... Protein

415. Which of the. following is the backbone of all organic matter?. .....Carbon

416. Which of the following is used as antimicrobial agent?..... All of the given options

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417. Which of the following are the first living organisms on our planet?. bacteria

418. . In anaerobic respiration, final electron acceptor is inorganic substance other than O2

419. is..... All of the given options

420. The man behind the cell theory WaS.....
Robert hooke

421. . The habitate of Escherichia coli ……..... Large intestine

Prokaryotes have no .,,,

Carbohydrate

437. _____ Directs the light through the specimen…..... Condenser

438. Duration of Golden age of Microbiology is: ……..1857 – 1914

439. Pasting of sample on glass slide is known as …….. Smear Formation

440. Monitoring diseases in a population and detection of outbreaks and developing epidemics are studied under the discipline of Epidemiology

441. The primary stain of Gram's method is – …….. Crystal Violet

442. Bacteria obtain nitrogen from air by the process of Nitrogen fixation

443. . Aerobes that require 02 in low amounts are called as …….. Microaerophiles

444. Helicobacter pylori is a…… Aerotolerant Aerobe

445. . oiled extracts of hay or meat can give rise to microorganisms, this statement proposed by ..... John Needham

446. the names of domains in three domain classification of microbes are…….. Bacteria, Archea and Eucarya

447. Which of the following is the physical source of growth?. All of the given options

448. . Living thing originate from a non-living thing is KNOWN as…….. None of the given options

449. . Which of the following initiate the spore activation Heating

450. Population of cells derived from a single cell is known as Clone

451. . Which one Is correct answer regarding sterilization…….. Removal of microbes and their spores

452. Five kingdom classification was proposed by the Carl Linnaeus

453. Antigen antibody reactions is a Methods for microbial identification… Serological

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454. Growth of medium in a liquid media with closed vessel is known as Batch culture.

455. . Ethylene oxide is strong alkylating agent that kills the microbes by reacting with

456. functional groups of block replication and enzymatic activity to DNA

457. . Which of the following is used to see antibody antigen reaction?..... Both radioisotopes and enzymes

458. . A nutrient material for growth is known as .....
Culture medium

459. Prokaryotes and Eukaryotes are .....
similar ..... Chemically

NADH and FADH2 during Kreb's cycle oxidized via Electron transport chain and

460. generate ATP molecules.....34

461. Which of the following | is required for treatment of diseases and 462.
cancer?..... antibodies

463. The term "animalcules" for microbes was firstly used by..... Antony van Leeuwenhoek

464. Peptidoglycan is made up of.....
Carbohydrates

465. . Microbes first drawing ..... published in... 1665

466. Which is the correct order of Acid Fast Staining ?..... Carbol Fuschin > Crystal Violet > Decolorizer > Methylene Blue

467. Which of the following compounds are needed for nucleic acid

468. synthesis?.....Sulfur and phosphorus

469. Which of the following is anoxygenic?.....
Rhodopseudomonas

470. Three domain classification was proposed by the Carl woese

471. Removal and ..... gain of electrons from an atom is known as Redox reactions

472. Which of the following stain is used as simple stain?.....
Methylene Blue

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473. Which of the following is acidophile?..... Both molds& yeast

474. What are the physical requirements of bacteria?. ..... All of the given option

475.. Microorganisms that have high nutritional requirements are known as ……

None of the given options

476. Which of the following compounds are needed for protein synthesis……

Nitrogen and Sulfur

477. Eyepiece is also known as Ocular Lens

478. Which of the following formed inside bacterium?.....

Endospore

479. Anthrax is caused by.....Bacillus anthracis

480. Phosphate ions ..... ATP

481. . Which of the following is not a bacterial gene transfer method?.....

Transfection

482. Which of the following is used for gene mapping?.....

Conjugation

483.. Which of the following is relevant to respiration of glucose?...All of the given options

484. Removing microbes from a limited area is known as.....

Degerming

485. Are live mostly in extrema conditions?.....Archea

486. Whic of the following staining technique is used as differential

487. staining?.....

Gram Staining

488. Biological oxidations involve the loss of hydrogen atoms, they are also called ..... reactions……

…… Oxidation

.486. Pasteur found the ..... alcohol that fermented sugar to ..... Yeast

487. Besides water, is the most important requirement for microbial growth…… Carbon

488. . Specialized resting cells .....are known as Endospores

Organism are classified into ..... kingdom in latest

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classification 5

489. Cold loving bacteria are known as ..... Psychrophiles

490. Which of the following is used to maintain the desired pH?. Buffers

491. . Eukaryotes have distinct as point of difference to the prokaryotes..... Nucleus

492. Who is first to publish the extensive and accurate observations of microbes?..... Antony van Leeuwenhoek

493. Which of the following is the reason for changes in the classification?..... All of the given options

494. Which of the following is the dye for positive staining?..... All of the given options

495. Which of the following is the dye for negative staining?. Eosin

496. The theory of spontaneous generation was proposed by Aristotle

497. The cell wall of bacteria is mostly made up of Peptidoglycan

498. For smear formation..... we use Platinum loop

The study of immune..... system is known as ..... Immunology

499. Which of the following is used for microbial control through chemicals?.....

Aldehydes

500. _____is very basic analytical technique that is used to separate solid insoluble Particles from liquid.....

Filtration

501. An extra chromosomal circular DNA fragment is Known as Plasmid

502. For which power of lens, ...we use oil X

503. Antigen _ Antibody reaction..... is known as Agglutination

504. Who discovered the first vaccine?.....

Edward Jenner

505. Which of the following is the illumination source for fluorescent

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506. microscope?..... Both Fluorochromes & UV light

. Oxidation of glucose to pyruvic acid with the prduction of some ATP and energy

507. containing NADH is .....known as .....Glycolysis

508. Microscopy that is required for unstained specimens is known as 509. .... Darkfield Microscopy

510. . NADH and FADH2 are oxidized during..... Electron transport chain

511. Mycology is branch of biological science which deals with the study of Fungi

512. The optimal temperature for growth of hyperthemophils is.....80 degree

513. Nucleoid and ribosomes are formed ..... Spore core

514. Immunology deals with the study of .....

ANS : Immune system

515. In gram positive bacteria layer of... ..... is thick.

ANS : peptidoglycan

516. The basics shapes of bacteria are : ANS : All of them

517. NADH and FADH2 during krebs cycle oxidized via electron transport chain and generate ATP molecule.

ANS : 38

518. Most of the bactreria grow at PH between .....

ANS : 6.5 7.5

519. Which of the statement is not correct for active movement . Ans. none of them

520. The cell membrane can be destructed by using .....

Ans: Alcohol

521. The drawing of microbes was first published in .....

Ans: 1665

522. Who is the founder of vaccinology Ans: Edward jenner

523. The microbes which contain ssRNA and need specific viruses to co- infect are called .....

Ans: virusoid

524. Which of the following are microbial products? Ans: All of them

525. According to the latest system of classification organism are classified into

Ans: 5

526. Nitrogen and phosphorus is required for the synthesis of .....

Ans: both of them

527. Smear can be faced by :

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Ans: heat

528. When cocci joins in chain like structure then the shape is known as …….

Ans: Streptococci

529. The microbes which need oxygen for their growth are called ? Ans: obligate anaerobes

530. Classification of mkcrobes is important to…….

Ans: all of these

531. In catalase test , hydrogenperoxide is converted into…….

Ans: water and oxygen

532. The bacteria which don't possess flagelum are called? Ans: Atrichous

533. The golden age of microbiology was spreadheaded by the discoveries of ……

Ans: both of them

534. Louis Pasteur used ……. In hisexperiment which helped him to end the concept of spontaneous generation.

Ans:

535. The main reason discoveredby Pasteur of souring of beer and wine was :

Ans:

536. Which of the following are too small to be seen with the unaided eye?

Ans: both germs & microorganism

537. Refractive index is the ……

Ans : Ability of a medium to bend the light

538. Which is the correct order of germ staining?

Ans: crystal violet > gram iodine > decolorizer > safranin

539. The study of pandemics endemics and epidemics is done in: Ans: Epidemiology

540. The movement of microorganismis response of light is called ……..

Ans: phototaxis

541. ……. are the reserved deposits of nutrients and avoid the increase in osmotic pressure .

Ans: Inclutions

542. A colony or a drop of broth culture can be picked by using …….

Ans. Platinum loop

543. The most common carbohydrates energy source in microbes is …….

Ans: glucose

544. ……. dyes are most commonly used for bacteria . Ans: basic

545. Chemoautotrophs and photoautotrophs use ……. As carbon source.

Ans. CO_2

546. Which of the following organism mostly live in extreme conditions ? Ans : Archea

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547. special staining include. Ans . all of these

548. which of the following is acidophiles? Ans . both of them

549. Hyperthermophiles lives in ……

Ans. Hot spring

550. What is the temperature require for sterilization in hot air oven ? Ans. 170 degree / not confirmed

551. Prokaryotes divide by…….

Ans: binary fission

552. The…….protein are imbedded in the membrane are amphipathic in nature and act as a receptors as well

Ans: Integral

553. The instrument used to see microbes is called ……..

Ans: Microscope

554. The unicellular form of fungi is known as ……

Ans : yeast

555. The basic shapes of archea are : Ans: all of these

556. Monitoring of disease in populations is known as ……

Ans : Epidemiology

557. Which one is correct answer regarding sterilization. Ans: removal of microbes and their spores

558. which of these is a type of differential staining? Ans: Gram staining

559. Half of the dry weight of microbe is ……

Ans: Phosphorous

560. The multicellular form of fungi is known as :…….

Ans : /mold

561. which of the following organism act as protozoa like at one stage and human like in another? Ans: smile mold

562. Who discovered the first

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vaccine ? Ans: Edward jenner

563. Which of the following equations is

correct ?\ Ans : 1mm=100µm

564. The..... do not use hydrogen from water but from H2S and h2 to reduce co2.

Ans : Photoautotrophs

565. organism lived in high salt concentration are known as .....

Ans: Halophiles

566. inflammation of tongue and mouth cavity (black tongue) caused by .....

Ans: Niacin

567. Decolorizing agent is used in gram

staining as : Ans: Alcohol

568. The heat loving bacteria are called .....

Ans: Thermofiles

569. Which of the following is used for plasma membrane

disruption ? Ans: c. Both phenol and phisphenol

570. which of the following statement is not true for fermentation?

Ans: Does not require

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571. Which of the following is the type of membraneprotein ? ANS: integral protein

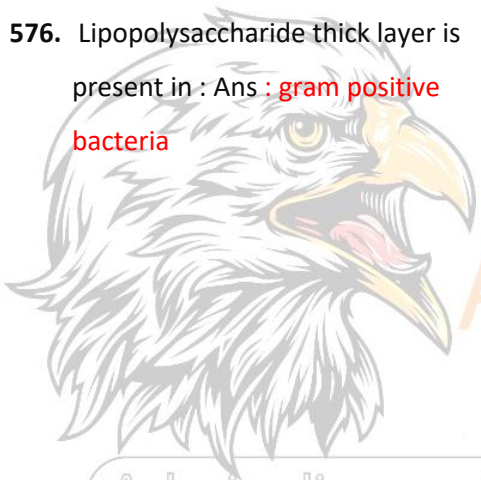
572. Which of the following equation is correct ? Ans: 1mm = 1000µm

573. The branch of mycology which deals with the study of decomposers of wood ,leaves and others organic matter . Ans: Ecological mycology

574. in scanning electron microscope a beam of electron is used to focus the ……. of cells. Ans : Surface of cells

575. oil is used for ……..
Ans: 100X

576. Lipopolysaccharide thick layer is present in : Ans : gram positive bacteria



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